Chapter 9 — Sentiment case study and big-data outlook

The chapter closes by tying advanced collection to an end-to-end application

Learning objectives

- Emerging trends such as edge and AI-driven analytics

Social platforms as continuous sentiment sources

- Platforms such as Twitter, Facebook
- Those signals can be analyzed for sentiment and related insights when collection

Example 9.28 — Real-Time Twitter Sentiment for Launch Feedback

- Example 9.28 — hands-on module
- Example 9.28 considers a product launch monitored through Twitter streams
- Visualize results on dashboards
- Explore the chapter example module
- View files: `modules/chapter9/example28/`

Distributed file systems and ingestion modes

- Scalable storage
- Ingestion may be batch
- Mode choice follows latency needs

Quality and privacy in big-data environments

- Duplicates that skew analysis
- Cleaning, transformation, and validation are therefore part of collection design
- Privacy requires encryption, access control

Integration with AI and machine learning

- Big data realizes more value when combined with AI and machine learning
- Predictive analytics in healthcare, finance, and retail forecast outcomes or demand
- Recommendation systems at scale use behavior histories to suggest products or media
- Collection pipelines must therefore deliver timely

Emerging trends and outlook

- Emerging directions include quantum computing for faster complex processing as systems
- These trends extend the advanced collection themes of this chapter

Takeaways

- Real-time sentiment analysis shows how streaming collection, distributed transport, NLP
- HDFS-style stores and batch or real-time ingestion land those streams
- Quality, privacy, AI integration

Next

- Complete the quiz for this part
- Complete the quiz for this part to close Chapter 9